Lab 9: Logistic and Count-Variable Regression

Python for Health Analytics | October 25, 2026

Logistic regression: analysis

1. What is the model accuracy? The accuracy is 0.7037, or **70.37%** (133 correct predictions out of 189).

2. What is the lowest precision? Class 1 (low birth weight) has the lowest precision: **0.56 (56%)**.

3. Do all categories have at least 30 support? Yes. Class 0 has **130** observations and class 1 has **59**. The confusion matrix is [[119, 11], [45, 14]].

4. Is this a good model based on pseudo-R^2? McFadden pseudo-R^2 is **0.0625**, suggesting modest explanatory power. Although overall accuracy is 70.37%, recall for class 1 is only **0.24**, meaning the model misses many low-birth-weight cases. It is not especially strong for identifying the positive class.

Poisson regression: analysis

1. Which four predictors are significant? Using $p < 0.05$: **depression_crudeprev** ($p < 0.001$), **selfcare_crudeprev** ($p = 0.007$), **indeplive_crudeprev** ($p < 0.001$), and **loneliness_crudeprev** ($p < 0.001$).

2. Which is most significant based on the coefficient? Of the four significant predictors, **selfcare_crudeprev** has the largest absolute coefficient (**-0.0608**), indicating the largest estimated coefficient magnitude per one-unit increase, holding the others constant. Among positive coefficients, **indeplive_crudeprev** is largest (+0.0522). If “most significant” means strongest statistical evidence rather than coefficient size, **depression_crudeprev** has the smallest p-value (approximately $6.64 \times 10^{■■}$). Coefficients are not directly comparable as importance measures when predictor scales differ.

3. Do any significant predictors have VIF > 10? Yes: **selfcare_crudeprev** (VIF = **30.51**) and **indeplive_crudeprev** (VIF = **30.95**). This indicates substantial multicollinearity and warrants caution when interpreting their individual coefficients.

4. How much variability is captured by the model? The saved output reports **deviance-based pseudo-R^2 = 0.9260 (92.60%)**. The GLM summary also reports **Cox–Snell pseudo-R^2 = 0.1925**. These are different measures; the 92.60% figure is the deviance-based value requested in the hint, not a conventional ordinary least-squares R^2.

Modeling note

The assignment specifies Poisson regression with mhlth_crudeprev as the response, although this is a crude prevalence measure rather than a raw count. Results should therefore be interpreted cautiously; Poisson count-model assumptions may not be appropriate for this outcome.

Full saved logistic regression output

```
LOGISTIC REGRESSION RESULTS
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                           Logit Regression Results
==============================================================================
Dep. Variable:                    low   No. Observations:                  189
Model:                          Logit   Df Residuals:                      185
Method:                           MLE   Df Model:                            3
```

```
Date:                Sun, 25 Oct 2026   Pseudo R-squ.:                0.06250
Time:                        14:21:16   Log-Likelihood:               -110.00
converged:                       True   LL-Null:                      -117.34
Covariance Type:            nonrobust   LLR p-value:                 0.002125
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.3551      0.241     -5.611      0.000      -1.828      -0.882
smoke          0.6831      0.329      2.075      0.038       0.038       1.329
ht             1.4174      0.624      2.273      0.023       0.195       2.640
ui             1.0384      0.428      2.426      0.015       0.200       1.877
==============================================================================

Confusion Matrix (0, 1):
[[119  11]
 [ 45  14]]

Classification Report:
              precision    recall  f1-score   support

           0       0.73      0.92      0.81       130
           1       0.56      0.24      0.33        59

    accuracy                           0.70       189
   macro avg       0.64      0.58      0.57       189
weighted avg       0.67      0.70      0.66       189

Accuracy: 0.7037
McFadden Pseudo R-squared: 0.0625
```

Full saved Poisson regression output

```
POISSON REGRESSION RESULTS
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                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:        mhlth_crudeprev   No. Observations:                 2299
Model:                            GLM   Df Residuals:                     2287
Model Family:                 Poisson   Df Model:                           11
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -5397.6
Date:                Sun, 25 Oct 2026   Deviance:                       39.316
Time:                        14:30:40   Pearson chi2:                     39.2
No. Iterations:                     4   Pseudo R-squ. (CS):             0.1925
Covariance Type:            nonrobust
===========================================================================================
                              coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------------
const                       1.7032      0.172      9.888      0.000       1.366       2.041
binge_crudeprev             0.0032      0.003      0.954      0.340      -0.003       0.010
checkup_crudeprev          -0.0004      0.002     -0.192      0.848      -0.004       0.004
csmoking_crudeprev          0.0043      0.004      1.100      0.271      -0.003       0.012
depression_crudeprev        0.0137      0.002      5.800      0.000       0.009       0.018
obesity_crudeprev           0.0001      0.002      0.076      0.940      -0.003       0.004
sleep_crudeprev             0.0024      0.002      1.059      0.289      -0.002       0.007
selfcare_crudeprev         -0.0608      0.022     -2.708      0.007      -0.105      -0.017
indeplive_crudeprev         0.0522      0.014      3.759      0.000       0.025       0.079
loneliness_crudeprev        0.0113      0.003      3.717      0.000       0.005       0.017
foodinsecu_crudeprev       -0.0083      0.007     -1.247      0.212      -0.021       0.005
housinsecu_crudeprev        0.0150      0.008      1.786      0.074      -0.001       0.032
===========================================================================================

Variance Inflation Factors:
           Predictor       VIF
     binge_crudeprev  2.528842
   checkup_crudeprev  1.750444
  csmoking_crudeprev  6.934794
depression_crudeprev  2.216376
   obesity_crudeprev  2.573017
     sleep_crudeprev  3.209334
  selfcare_crudeprev 30.514413
 indeplive_crudeprev 30.953309
loneliness_crudeprev  3.432565
foodinsecu_crudeprev 62.830787
```

```
housinsecu_crudeprev 46.839954

Significant Predictors (p < 0.05):
depression_crudeprev    6.637099e-09
selfcare_crudeprev      6.769180e-03
indeplive_crudeprev     1.706267e-04
loneliness_crudeprev    2.014381e-04

Deviance-based Pseudo R-squared: 0.9260
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